

Neeraja Santhosh Kumar

Integrated M.Sc. Student, Biology · NISER Bhubaneswar

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About Me

I am a second-year Integrated M.Sc. student at NISER Bhubaneswar interested in Quantitative Biology and Biophysics. I am drawn to problems where dynamical systems, computation, and biological structure intersect and hope to build a deeper expertise at this interface.

Education

National Institute of Science Education and Research (NISER)

Aug 2025 – Present

Integrated M.Sc., Biological Sciences | Second Year | Bhubaneswar, Odisha

Chinmaya Vidyalaya

Kannur, Kerala

- Class XII (CBSE): 93.6% (2023) | Class X (CBSE): 96.9% (2021)

Research Experience

Research Intern, Synchronization of Adaptive Dynamical Networks

May 2026 – Present

Complex Systems Lab, IISER Thiruvananthapuram | Supervisor: Dr. D.V. Senthilkumar

- Investigating synchronization phenomena in adaptive dynamical networks as part of an ongoing research project; continued remotely following the on-site internship period.

Research Intern

Dec 2025

Biogeosystems Lab, NISER Bhubaneswar | Supervisor: Dr. Aniruddha Datta-Roy

- Used Python and ImageJ to perform morphometric analysis and study the optics of animal camouflage, an intersection of biological structure and physical measurement.
- Compiled a comprehensive presentation on the mechanics of camouflage in ground-dwelling birds of India.

Workshops, Schools & Conferences

Summer School for Women in Mathematics and Statistics

2026

ICTS-TIFR, Bengaluru

- Completed a rigorous, selective two-week program covering linear algebra, probability, logic, and dynamical systems, with daily problem sessions, R-based simulations, and talks by researchers.

Conference on Nonlinear Systems and Dynamics (CNSD)

Jul 2026

IISER Thiruvananthapuram | Volunteer

- Assisted in the efficient organization of the conference as part of the Media team and attended talks by eminent researchers.

Winter School, Centre for Nano Science and Engineering (CeNSE)

Dec 2025

IISc Bangalore

- Attended winter school program focusing on nanofabrication.
- Gained exposure to micro-fabrication techniques and Finite Element Analysis (COMSOL) for MEMS applications.

Skills

- **Programming & Computation:** Python (NumPy, SciPy), C, C++, R, MySQL, L^AT_EX
- **Imaging & Version Control:** ImageJ/Fiji (bio-image analysis), Git
- **Web Development:** Django, JavaScript
- **Languages:** English (Fluent), Hindi (Fluent), Malayalam (Fluent), Tamil (Intermediate)

Academic Achievements

VSSF SPOT 100 GENII

2021

Vikram Sarabhai Space Foundation

- National Top 100 qualifier; selected for the senior-level exam and mentorship sessions with eminent researchers.

- Secured State Rank 1 in the VVM national talent search examination; participated in state and national camps with hands-on problem-solving activities.

Leadership & Extracurricular Activities

- **Core Member, TheBioClub NISER:** Develop content and contribute to club initiatives.
- **Content Editor, NISER Astronomy Club:** Curate scientific outreach material and club content.
- **Community Ambassador, Write the World** (Jun 2021 – Sep 2021): Conducted peer reviews for published writing, developed writing prompts and exemplars for the community, and collaborated with an international team of ambassadors in weekly strategy sessions.
- **Senior Editor, Polyphony Lit Magazine** (2020–23): Mentored junior editors of an international literary magazine and managed content workflows.
- **Greenlight4girls Ambassador** (2020–23): Organized events to encourage girls in STEM and interacted with international ambassadors.
- **Assistant School Leader** (2021–22): Led student council initiatives and coordinated school events.
- **Quiz Team Member** (2021–23): Won Sahodaya Championships for Science and General Knowledge in North Kerala.